

BPHO ROUND 0 · QUESTION PAPER

Drill bank section 7

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking

Original questions written for this course, in the Round 0 format. Not BPhO material.

Before you start

- Answer every question. There is no negative marking, so a blank is a wasted mark.
- No calculator. Every answer here is reachable by hand, and the numbers are chosen to divide cleanly.
- Work on paper, then mark yourself against the markscheme PDF.
- The 2025 UK qualifying line was 11 out of 25, so about 44% is the target to beat.

Q1 · Dimensional analysis by solving for the exponents: the combination of h , c and G that has the dimensions of a length

Three constants of nature have these dimensions:

the Planck constant h , of dimensions $\text{kg m}^2 \text{s}^{-1}$; the speed of light c , of dimensions m s^{-1} ; and the gravitational constant G , of dimensions $\text{m}^3 \text{kg}^{-1} \text{s}^{-2}$.

Which combination of them has the dimensions of a length?

A $\sqrt{(hc/G)}$

B $\sqrt{(hG/c)}$

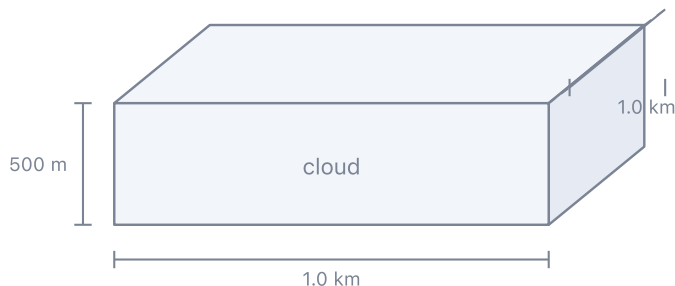
C $\sqrt{(hG/c^3)}$

D $\sqrt{(hc^3/G)}$

E $\sqrt{(hGc)}$

Q2 · Order-of-magnitude estimation: the mass of water in a cloud, from its volume and its liquid water content A

A cloud is roughly a slab 1.0 km long, 1.0 km wide and 500 m deep, as the diagram shows. The liquid water content of a cloud of this kind is about 0.50 g m^{-3} .



Roughly what mass of water does the cloud contain?

- | | |
|---------------------------------------|---------------------------------------|
| A $2.5 \times 10^5 \text{ kg}$ | B $2.5 \times 10^8 \text{ kg}$ |
| C $2.5 \times 10^2 \text{ kg}$ | D $2.5 \times 10^6 \text{ kg}$ |
| E $5.0 \times 10^5 \text{ kg}$ | |
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Q3 · A small fractional error through a cube: the radius of a sphere is measured one per cent too large, so the volume is three per cent too large A

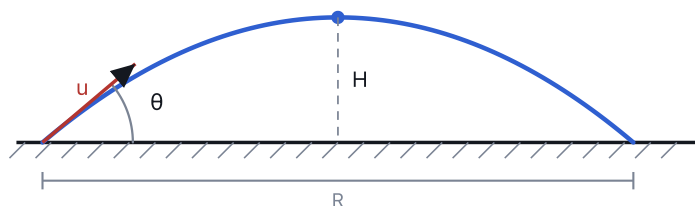
The radius of a sphere is measured and found to be **1.0% too large**. The error is in the measurement of the radius only.

By roughly what percentage is the volume calculated from it too large?

- | | |
|----------------|---------------|
| A 0.33% | B 1.0% |
| C 2.0% | D 3.0% |
| E 9.0% | |
-

Q4 · A projectile: the ratio of its maximum height to its horizontal range, which depends on the launch angle alone **B**

A ball is projected from level ground with speed u at an angle θ above the horizontal, and lands at the same height, as the diagram shows.



Which expression gives the ratio of its maximum height to its horizontal range?

A $\tan\theta / 2$

B $\tan\theta / 4$

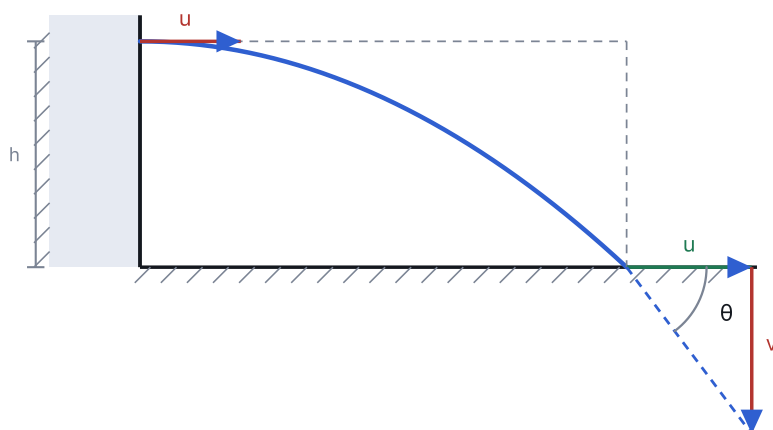
C $\tan\theta$

D $2 \tan\theta$

E $4 \tan\theta$

Q5 · A projectile launched horizontally from a height: the tangent of the angle at which it lands **B**

A ball is thrown horizontally with speed u from the top of a cliff of height h , and strikes the ground below. The acceleration due to gravity is g .



Which expression gives the tangent of the angle between the ball's velocity and the horizontal at the moment it lands?

A $u / \sqrt{2gh}$

B $2gh / u$

C $\sqrt{gh} / u$

D $2 \sqrt{gh} / u$

E $\sqrt{2gh} / u$

Q6 · A perfectly inelastic collision with a stationary target three times as heavy: the fraction of the kinetic energy lost

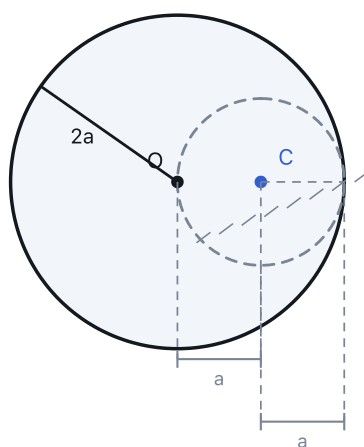
A particle of mass m travelling at speed u strikes a stationary particle of mass $3m$ and the two stick together.

What fraction of the original kinetic energy is lost in the collision?

- ☐ A $3/4$
☐ B $1/4$
☐ C $1/2$
☐ D $2/3$
☐ E $3/8$
-

Q7 · A uniform disc with an off-centre circular hole: where its centre of mass sits, treated as a negative mass

A uniform circular disc of radius $2a$ has a circular hole of radius a cut out of it. The centre of the hole is a distance a from the centre of the disc, as the diagram shows.



Where is the centre of mass of the remaining shape?

- ☐ A A distance $a/2$ from the centre, on the same side as the hole
☐ B A distance $a/4$ from the centre, on the side away from the hole
☐ C A distance $a/3$ from the centre, on the side away from the hole
☐ D A distance $a/3$ from the centre, on the same side as the hole
☐ E A distance $2a/3$ from the centre, on the side away from the hole
-

Q8 · Terminal speed of a falling body when the drag is proportional to the square of the speed **C**

A body of mass m falls through a fluid. The drag force on it is $k v^2$, where v is its speed and k is a constant.

Which expression gives its terminal speed?

A mg / k

B $k / (mg)$

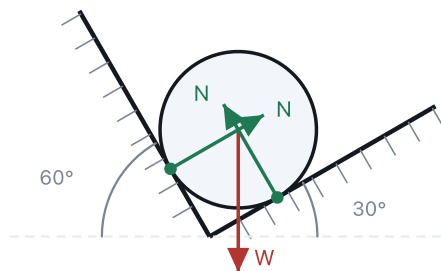
C $\sqrt{k / (mg)}$

D mg / k^2

E $\sqrt{mg / k}$

Q9 · A cylinder resting in a V-groove with faces at 60 degrees and 30 degrees: the force on the steeper face, which is the smaller of the two **C**

A uniform cylinder of weight W rests in a V-groove. The left face of the groove is inclined at 60° to the horizontal and the right face at 30° , as the diagram shows. Both faces are smooth.



Which expression gives the force exerted by the left face?

A $W / \sqrt{3}$

B $W / 2$

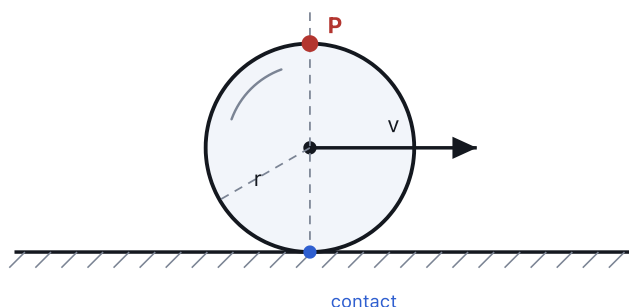
C $\sqrt{3} W / 2$

D W

E $2 W / \sqrt{3}$

Q10 · A wheel rolling at constant speed: the acceleration of the point at the top of the rim, which is the same magnitude as at every other point ☐ D

A wheel of radius r rolls along a level road without slipping, at a constant speed v .



What is the magnitude of the acceleration of the point at the top of the wheel?

☐ A 0

☐ B $2v^2 / r$

☐ C $v^2 / (2r)$

☐ D v^2 / r

☐ E $4v^2 / r$

Q11 · A rod clamped at both ends and heated: the thermal stress, which is the stress needed to undo the expansion that could not happen ☐ E

A metal rod is clamped firmly at both ends so that it cannot change length, and its temperature is then raised by ΔT . The metal has linear expansivity α and Young's modulus E .

Which expression gives the stress that appears in the rod?

☐ A $E \Delta T / \alpha$

☐ B $\alpha \Delta T / E$

☐ C $E \alpha / \Delta T$

☐ D $E \alpha \Delta T^2$

☐ E $E \alpha \Delta T$

Q12 · The speed of sound in a gas compressed at constant temperature: it does not change, because the density rises in step with the pressure ☐ F

The speed of sound in a gas is given by $v = \sqrt{(\gamma P / \rho)}$, where γ is a constant for the gas, P is its pressure and ρ is its density.

A fixed mass of the gas is compressed **at constant temperature** until its pressure is twice what it was. What happens to the speed of sound in it?

☐ A It doubles.

☐ B It is unchanged.

☐ C It is halved.

☐ D It rises by a factor of $\sqrt{2}$.

☐ E It falls by a factor of $\sqrt{2}$.

Q13 · Beats between two tuning forks: the direction the beat rate moves when one fork is loaded tells you which side of the known frequency the unknown lies on **F**

A tuning fork of frequency 512 Hz is sounded beside a second fork of unknown frequency. **Four** beats per second are heard.

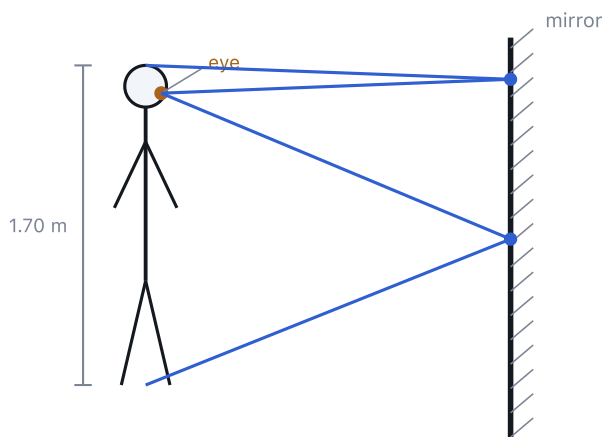
Wax is then put on the 512 Hz fork, which lowers its frequency, and the beat rate falls to **two** per second.

What is the frequency of the second fork?

- A** 508 Hz
 - B** 516 Hz
 - C** 510 Hz
 - D** 514 Hz
 - E** 520 Hz
-

Q14 · The least height of a plane mirror in which a person can see their whole reflection, which is half their height and independent of the distance **G**

A person of height 1.70 m stands in front of a vertical plane mirror and wants to see their whole body, from the top of their head to the soles of their feet.



What is the least height of mirror that will do this?

- A** 1.70 m
 - B** 0.43 m
 - C** 0.85 m
 - D** 0.57 m
 - E** It depends on how far the person stands from the mirror.
-

Q15 · The smallest angle a telescope objective can resolve, which is set by the ratio of the wavelength to the diameter **G**

The smallest angle that a telescope objective of diameter **D** can just resolve is about λ/D , where λ is the wavelength of the light.

A telescope has an objective 0.10 m in diameter and is used with light of wavelength 5.5×10^{-7} m. What is the smallest angle it can just resolve?

- A** 5.5×10^{-7} rad **B** 1.1×10^{-5} rad
C 2.8×10^{-6} rad **D** 5.5×10^{-6} rad
E 5.5×10^{-5} rad
-

Q16 · The cost of running a heater, built as a chain of rates: power, then hours, then days, then a price per kilowatt-hour **H**

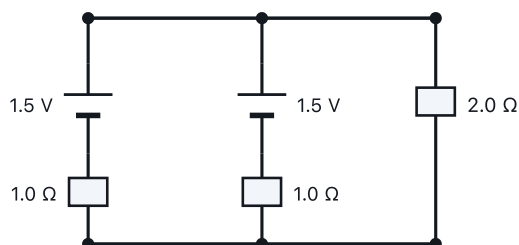
A heater is rated at 2.0 kW and is used for about 3.0 hours a day. Electricity costs 15 p per kilowatt-hour.

Roughly what does it cost to run for 30 days?

- A** £9 **B** £27
C £54 **D** £90
E £270
-

Q17 · Two identical cells in parallel driving a resistor: the e.m.f. stays at one cell's value while the internal resistance halves **H**

Two identical cells are connected in parallel. Each has an electromotive force of 1.5 V and an internal resistance of 1.0Ω . The combination is connected across a resistor of 2.0Ω , as the diagram shows.



What current flows in the 2.0Ω resistor?

- A** 1.5 A **B** 0.50 A
C 0.75 A **D** 0.60 A
E 0.30 A
-

Q20 · A lead ball dropped from a height: the rise in temperature, in which the mass cancels and only the height and the specific heat capacity matter

A lead ball is dropped from rest from a height of 20 m onto a hard surface. Assume that all of the energy it has on impact goes into internal energy in the ball.

Lead has a specific heat capacity of $130 \text{ J kg}^{-1} \text{ K}^{-1}$, and g may be taken as 10 m s^{-2} . Roughly what is the rise in the ball's temperature?

A 0.15 K

B 1.0 K

C 1.5 K

D 2.0 K

E 15 K

Q21 · A sealed tyre warmed by a journey: the gauge pressure rises by a larger fraction than the temperature, because the gauge shows only the excess over the atmosphere

A car tyre is inflated on a cold morning, when the air inside it is at 20°C , until the pressure gauge reads 2.0 atm. After a long drive the air inside has warmed to 50°C . The volume of the tyre has not changed.

Atmospheric pressure is 1.0 atm. Roughly what does the gauge read now?

A 1.7 atm

B 2.0 atm

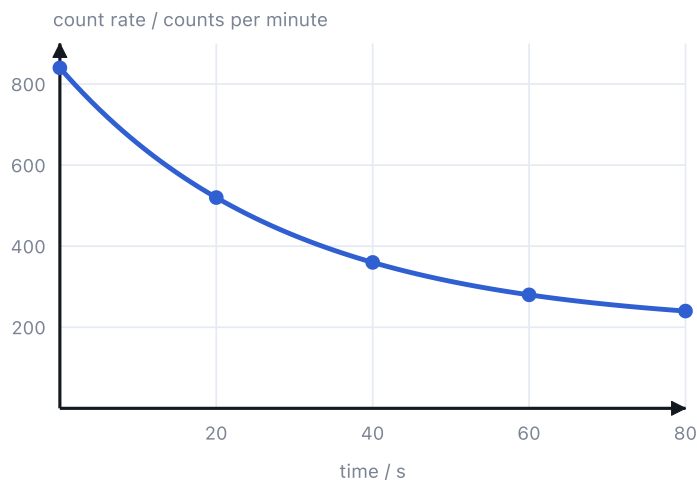
C 2.2 atm

D 2.3 atm

E 3.3 atm

Q22 · A decay curve recorded with a steady background: the background must be subtracted before the curve can be read, and only then does it halve K

A radioactive source is placed close to a detector, and the count rate recorded by the detector is plotted against time. The laboratory has a steady background count rate of 200 counts per minute from other sources, and this does not change while the measurement is made.



Roughly what is the half-life of the source?

- A** 10 s
- B** 20 s
- C** 30 s
- D** 40 s
- E** 80 s

Q23 · The energy released by one fission, estimated from the rise in binding energy per nucleon rather than from a remembered figure K

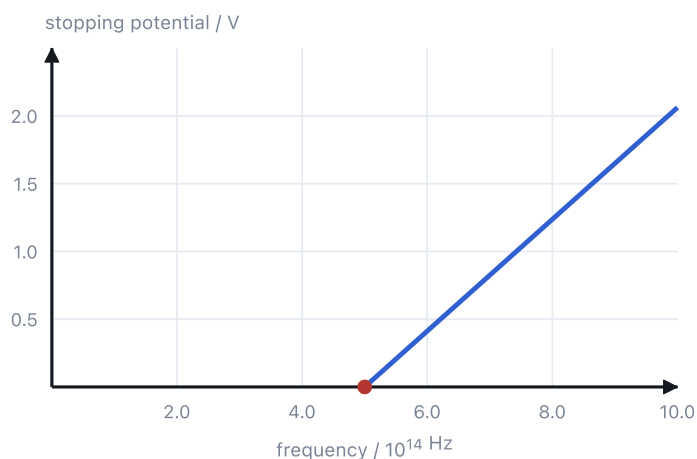
When a heavy nucleus of mass number 236 splits into two fragments, the binding energy per nucleon rises from 7.6 MeV in the original nucleus to 8.5 MeV in the fragments.

Roughly how much energy is released by the fission of one such nucleus?

- A** 2.1 MeV
- B** 21 MeV
- C** 210 MeV
- D** 240 MeV
- E** 420 MeV

Q24 · The photoelectric effect across three steps: the threshold frequency read from a stopping-potential graph gives the work function, and the work function then gives the kinetic energy for a stated wavelength **L**

Light of wavelength 330 nm is incident on a metal surface. The graph shows how the stopping potential for the electrons emitted from this surface varies with the frequency of the incident light.



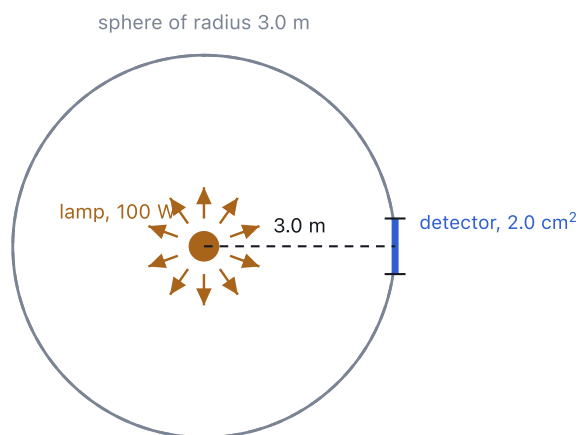
Take the Planck constant to be 6.6×10^{-34} J s, the speed of light in a vacuum to be 3.0×10^8 m s⁻¹, and the elementary charge to be 1.6×10^{-19} C.

Roughly what is the maximum kinetic energy of the emitted electrons?

- A** 0 eV
- B** 1.3 eV
- C** 1.7 eV
- D** 2.1 eV
- E** 3.8 eV

Q25 · Counting photons: the energy of one, the rate a lamp emits them at, and the fraction of a sphere that a small detector covers at a stated distance L

A lamp is rated at 100 W and radiates 5.0% of that power as light of a single mean wavelength of 550 nm. It may be treated as a point source radiating equally in all directions.



A small detector of area 2.0 cm^2 faces the lamp at a distance of 3.0 m.

Take the Planck constant to be $6.6 \times 10^{-34} \text{ J s}$ and the speed of light in a vacuum to be $3.0 \times 10^8 \text{ m s}^{-1}$. Roughly how many photons of the light strike the detector each second?

A 2.5×10^9

B 2.5×10^{13}

C 5.0×10^{14}

D 2.5×10^{15}

E 2.5×10^{17}